



Risk design optimization using many-objective evolutionary algorithm with application to performance-based wind engineering of tall buildings



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ABSTRACT

Risk design optimization (RDO) is a competent approach for automated performance-based structural design by achieving a balance between safety and economy. Performance-based wind engineering (PBWE) is aimed at improving the life-cycle functionality of wind-sensitive structures, hence could be the very field RDO is tailor-made for. In this paper, we embed PBWE of tall buildings into RDO and tackle some difficulties when integrating them directly. We first formulate an integrated stiffness and vibration control RDO problem, and employ a frequency domain closed-form solution for uncertainty quantification and uncertainty propagation through the excitation–response–performance chain. Then we reveal the multi-objective optimization nature of RDO, and circumvent the difficulties in serviceability loss estimation by replacing scalar total cost with high-dimensional objective vector. Micro multi-objective particle swarm optimization in conjunction with kernel-learning based principle component analysis is employed to solve the corresponding many-objective problem with multiple probabilistic constraints and discrete design variables. The optimization results of CAARC benchmark indicate that we simplify risk-based PBWE of tall buildings from a complex multi-objective decision making process into a relatively easy multi-attribute decision making process. Accordingly, convincing decisions can be made based on the explicit building performance rather than the unreliable loss information.

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1. Introduction

Performance-based design (PBD) improves risk assessment by introducing convincing scientific techniques, which enable stakeholders to make informed decisions. Extensive applications reveal its trend towards mitigating short duration hazards (e.g., earthquake [1,2], gust [3] and fire [4]) and long term effects, such as corrosion [5] and fatigue [6]. Among them, wind hazard in itself consists of multi-level hazard events, which correspond to multiple performance requirements. Naturally, performance-based wind engineering (PBWE) framework [3] is modified for long-span bridges [7] and building structures [8–10]. However, there still exists ample room for its improvement and one of the foremost requirements is to equip PBWE with optimization techniques [11].

The development of optimal PBD is directly related with the state of the art of structural optimization techniques. Hence wind-resistant design optimization for tall buildings has experienced a rapid development last several years. Chan et al. improved

the optimality criterion based stiffness optimization technique [12], and extended it to drift design [13] and habitability design [14,15] considering three-dimensional (3D) modes. Though these formulations were strictly derived, they cannot be expected to satisfy prescribed performance requirement. In the context of PBD, objective criteria are suggested to be described in a probabilistic manner, as the significance of any optimal solution would be relatively limited when the consideration of uncertainty is waived. Besides, to strictly enforce the performance objective, the target reliability should be set directly for the limit state [16] rather than the exceedance probability of mean wind recurrence interval [17].

Earlier contributions placed significance on the necessity to introduce probabilistic approaches in wind engineering [18–20]. Subsequently, great emphases were laid on the role of uncertainties in fundamental frequency and damping ratio for serviceability evaluation, because they may spread and shift the probability density function (PDF) of the resultant response [21–23]. The extent of the spreading would be amplified when multi-source uncertainties are arising through the excitation–response–performance chain. Reliability-based design optimization (RBDO) takes uncertainties into account and is under constraints related to the minimum

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target reliabilities specified by predefined performance requirements [24]. Applications of RBDO to structural design pose great challenge due to the large number of design variables and computational burden incurred from repeated reliability analyses. Approximate reliability method [25] and sampling-based method in conjunction with surrogate model [26–28] have been exploited to tackle these difficulties, and RBDO has shown competence on the design of wind-excited structure [29].

For the special issue on wind-resistant design of tall buildings, Spence et al. made original contributions by optimizing a tall steel frame under different fragility level [30,31], and extended the methodology into data-driven cases [32]. Huang et al. further improved the method in [13,15] by considering uncertainties in estimated wind speed and structural dynamic characteristics [33]. Though these RBDO approaches establish sound bases for automated PBWE of tall buildings, it has been empirically proved the optimality of RBDO solution fails when the structure would experience unavoidable damage in the future [34]. By striking a balance between safety and economy, risk design optimization (RDO) could be a competent alternative for PBWE of tall buildings. Its extensive applications to performance-based seismic design [35–37], an early attempt on mitigating hazard on wind-excited structure [38] and the formal propounding of risk-based PBWE methodology [7–10] indicate the possible benefit of integrating PBWE with RDO for tall building design. Owing to the uniqueness of tall building PBWE, existing probabilistic optimization approaches could be inappropriate to be directly extended into risk-based cases.

To bridge the gap between PBWE and RDO, PBWE of tall buildings is first formulated as an integrated stiffness and vibration control RDO problem in Section 2. Then we introduce 3D closed-form solution for uncertainty quantification and uncertainty propagation in Section 3. In Section 4, the original RDO problem is transformed into a many-objective RBDO problem. To solve this problem, we develop a constrained micro multi-objective particle swarm optimization (Micro-MOPSO) assisted with principle component analysis (PCA) in Section 5. Design optimization of a 45-story steel moment resisting frame example is performed in Section 6. Finally, we outline the conclusion of this paper in Section 7.

2. RDO formulation for serviceability PBWE of tall buildings

PBWE is originally proposed towards both ultimate limit state and serviceability limit state. Since wind-induced vibration may cause occupant discomfort, malfunction of facilities and excessive inter-story drift hence damage to non-structural elements (NSEs) [39,40]. Serviceability limit states play fundamental roles for tall buildings, especially the ones located in non-hurricane regions. All the failure consequences and the associated risks (expected loss) can be mitigated at the expense of introducing more initial safeties. From the structural design aspect, sizing optimization should be applied to all the structural elements for desired overall stiffness distribution. However, modifying the stiffness alone is low efficient to meet habitability performance requirement [14] as acceleration response is insensitive to the changes in structural frequencies and mode shapes. Hence we implement a smart tuned mass damper (STMD) at the top of the building to directly restraint the acceleration response. The RDO formulation for serviceability PBWE of tall buildings is formulated as

$$\begin{aligned} \text{Minimize}_x : C_T = & w_1 \left(C_U \sum_{i=1}^L (\rho_i l_i x_i) + C_D \right) + w_2 \left(\sum_{j=1}^{N+1} p_j C_{1j} \right) \\ & + w_3 \left(\sum_{k=1}^2 p_k C_{2k} C_L \right) \end{aligned} \quad (1)$$

$$\begin{aligned} \text{subject to : } & p \left(\frac{U}{H} \geq \delta_{\text{Top}} \right) \leq p_D^T; \quad p \left(\frac{V}{H} \geq \delta_{\text{Top}} \right) \leq p_D^T; \\ & p \left(\frac{\sqrt{U^2 + V^2}}{H} \geq \delta_{\text{Top}} \right) \leq p_D^T \end{aligned} \quad (2)$$

$$\begin{aligned} & p \left[\frac{u_j - u_{j-1}}{h} \geq \delta_j \right] \leq p_j^T; \\ & p \left[\frac{v_j - v_{j-1}}{h} \geq \delta_j \right] \leq p_j^T; \\ & p \left[\frac{\sqrt{(u_j - u_{j-1})^2 + (v_j - v_{j-1})^2}}{h} \geq \delta_j \right] \leq p_j^T, \quad (j = 1, \dots, N) \end{aligned} \quad (3)$$

$$\begin{aligned} & p[(1 - \gamma)a_{\text{Peak}} \geq a_{\text{Peak}}^T] \leq p_{\text{Peak}}^T; \quad p[(1 - \gamma)\sigma_\alpha \geq \sigma_\alpha^T] \leq p_{\text{RMS}}^T, \\ & \alpha = x, y, \theta \end{aligned} \quad (4)$$

$$\mathbf{x} \in \mathbf{X} \subset D^L \quad (5)$$

The first term of the weighted-sum objective function is the aggregation of material cost (with the i th material density ρ_i , element length l_i , cross-sectional area x_i and unit cost C_U) and STMD cost C_D as

$$C_D = (16.1m^* + 1.9)\gamma^2 - (6.8m^* + 1.7)\gamma + (1.5m^* + 2.2) \quad (6)$$

where m^* and γ are the first-order generalized mass and designated acceleration reduction level (ARL) [41], respectively. For example, when m^* ranges from 1.6 to 2.4 ($\times 10^7$ kg), C_D ranges from 3.99 to 5.08 (Million USD) conditioned under $\gamma = 0.4$. The coefficients in Eq. (6) vary slightly for different structural system, but the expression does involve the most essential factors that influence the total cost of STMD. Therefore, it is accurate enough to guarantee an optimal balance between structural member cost and STMD cost. Note that γ can be either defined as design variable or fixed.

Then we use a hierarchical serviceability PBWE framework to envelop the multi-level performance requirements [42]. Generally, there are two types of risk modeling approaches in PBWE. One approach is the widely employed design wind speed approach, in which the probability of failure (PF) and the corresponding risk are the functions of design wind speed. Another “fully-probabilistic approach” [10] convolves the PFs conditioned under mean wind speed with the PDF of mean wind speed, and derives the overall PF over time with a failure recurrence interval [8]. To keep consistency with conventional wind-resistant design concepts, we model the risks in the former way, and the latter approach could be very promising for life-cycle cost modeling.

Then each performance requirement is characterized with a mean wind speed and recurrence interval. Violation of elastic drift criteria indicates damage on NSEs and corresponds to design wind speed with a return period of 50 years. Probabilistic constraints in (2) and (3) define the maximum PFs (6.68%) [43] of p_D and p_j for atop deflection and inter-story drift ratio with peak thresholds $\delta_{\text{Top}} = 1/500$ and $\delta_j = 1/400$ (H and h are the building and story height), respectively. The corresponding three constraints are imposed on the along-wind response U , crosswind response V and response in the most unfavorable direction, respectively. Habitability performances are defined according to the types of loss consequences [44]. When performance loss is incurred from the violation of vibration tolerance and occupant discomfort, frequency-related root mean square (RMS) threshold is preferable [45] and a 5-year-recurrence design wind speed is utilized. In severe cases, occupant perception of floor vibration would bring fear, alarm and anxiety at the very instance, and interruptions in business would result in significant loss, so a peak value threshold is imposed with a recurrence interval of 10 years [46]. Very frequent wind with 1-year-recurrence interval is related to the daily

motion perception and the corresponding frequency dependent criteria are proposed in [47]. These habitability performances are constrained in (4), and the atop peak and RMS acceleration responses in sway and torsional directions are conditioned under the ARL γ . The maximum PFs for all the acceleration criteria are imposed as 50% due to their reversibility [43]. Eq. (5) defines an L -dimensional design space, in which D denotes a discrete set for each of the discrete variables.

The risks associated with the above performance loss should also be modeled in objective function (1). In serviceability PBWE, the damage on structural elements is trivial, so the risk is modeled as the overall expected loss consisting of NSE loss and business loss. We drop discount rate for illustration and introduce two sets of coefficients C_{1j} and C_{2k} to describe the reference loss associated with NSE and business downtime. p_j and p_k are the corresponding PFs, respectively. Additional transformation coefficient C_L is introduced to handle non-economic loss [42]. Risk-taking preferences [35] w_1 , w_2 and w_3 can be introduced to emphasize the minimization of different components. Habitability performance pertaining to 1-year-recurrence wind speed is imposed in the constraints, but the corresponding loss is negligible. All the failure modes are considered in series, thus the violation of any constraint would lead to an infeasible design during the design cycle. The most unfavorable direction of each performance is considered, though there are two orthogonal incident wind directions. Consequently, the total number of failure modes in the objective function is $N + 3$, i.e., N for inter-story drift ratio, one for atop deflection and two for habitability performances (namely the most unfavorable peak value and RMS value).

3. Uncertainty quantification and propagation using closed-form solution

When the probabilistic responses are derived in the above RDO formulation, uncertainty quantification and propagation through the excitation–response–performance chain play central roles. We employ a 3D closed-form solution [48–50] for dynamic analysis. As opposed to static wind load combination and equivalent static wind load, it takes the frequency contents of both excitation and structure into account. The quasi-steady model is accurate enough to capture the dynamic characteristics of the building under stochastic wind loads, and much more efficient than time history excitation when considering basic variable uncertainties, especially those in wind speed and aerodynamic force coefficients. When more rigorous description of body-initiated and signature turbulence is required, reduced order methods based on proper orthogonal decomposition could be applied to incorporate the wind tunnel determined stochastic wind loads without any particular loss of efficiency [32,51].

For a tall building with symmetric plan, wind-induced response of this mechanically uncoupled system can be evaluated in two orthogonal fundamental sway modes, i.e., those in the along-wind and crosswind direction, and the fundamental torsional mode independently. Reduced power spectra of turbulence excitation in along-wind and crosswind direction and wake flow excitation are respectively expressed as

$$\frac{fS_u(f)}{\sigma_u^2} = \frac{4f_u^2}{6(1+f_u^2)^{4/3}}; \quad f_u = \frac{1200f}{U_{10}} \quad (7)$$

$$\frac{fS_v(f)}{\sigma_v^2} = \frac{4f_v^2}{6(1+f_v^2)^{4/3}}; \quad f_v = \frac{800f}{U_{10}} \quad (8)$$

$$\frac{fS_s(f)}{\sigma_s^2} = \frac{A_s B_s (f/f_s)^{2+C_s}}{(1-f^2/f_s^2)^2 + B_s f^2/f_s^2}; \quad A_s = 3.5, B_s = 0.028, C_s = 1.4 \quad (9)$$

where $f_s = S_t U(z)/B(z)$ is the vortex shedding frequency, and S_t , $U(z)$ and $B(z)$ are the Strouhal number, mean wind speed and characteristic width at the height z , respectively.

These spectra are not deterministic in practice. For example, the mean wind speed at 10 m height U_{10} is conventionally derived using an extreme wind speed distribution, which is fitted based on site specified wind speed data. Generally, if the parameters describing a random variable are controllable, it can be imposed with a distribution with parametric uncertainty. For instance, a normal distribution has two parametric uncertainty sources, i.e., mean and standard deviation. On the part of design wind speed, Gumbel distribution has been approved to be an appropriate model.

$$X(n) = \mu - \sigma \ln \left[-\ln \left(1 - \frac{1}{n} \right) \right] \quad (10)$$

in which

$$\mu = E(X) - 0.45s(X) \quad (11)$$

$$\sigma = 0.78s(X) \quad (12)$$

Conditioned under the recurrence period n , the stochastic nature of design wind speed $X(n)$ arises from the location parameter μ and scale parameter σ (more essentially, the estimators of $E(X)$ and $s(X)$) due to limited samples. We obtain the statistics using point estimation [52]. Annual (50 samples) and daily (365 samples) maximum wind speeds are obtained from [53] with an average time of 3600 s at the reference height of 10 m. Particularly, day-to-day data are utilized to fit the parameters of daily extreme value distribution for 1-year-recurrence interval wind, while annual maximum wind speeds are used for the other cases. To keep consistency with the observation time in the wind tunnel tests [50], the original wind speed data are rescaled with an average time of 600 s [54]. In this way, statistical uncertainties owing to limited samples are modeled as normally distributed random variables (Table 1). Combined with the uncertainty in exposure constant, they would bring the variability of the wind speed spectrum (Fig. 1). Then the nominal values of design wind speeds at the top of the building are 29.6 m/s, 30.5 m/s, 34.0 m/s and 40.6 m/s corresponding to the recurrence intervals of 1, 5, 10 and 50 years, respectively.

Due to the computational burden posed by repeated structural eigenvalue analyses during optimization, the discretization degree of the excitation is set equal to the number of the floor levels hence there is a random process acting on each floor. Then the vertical coherence functions of height z_1 and z_2 with respect to fluctuating component and vortex shedding are expressed in (13) and (14), respectively.

$$Coh_e(z_1, z_2, f) = \exp \left(-\frac{fC_z|z_1 - z_2|}{\bar{U}} \right), \quad \varepsilon = u, v \quad (13)$$

$$Coh_s(z_1, z_2) = \exp \left(-\frac{|z_1 - z_2|}{HL_w} \right) \quad (14)$$

where C_z , $\bar{U}$, H and L_w denote the exponential decay coefficient, the average of the mean wind speed between z_1 and z_2 , overall height of the building and the ratio of wake flow turbulence scale versus building characteristic scale ($L_w = 1$ in this study). Based on quasi-steady theorem, generalized aerodynamic coefficients are applied to relating wind speed and vortex shedding cross-spectra

$$S_{ee}(z_1, z_2, f) = \frac{fS_e(f)}{\sigma_e^2} \times Coh_e(z_1, z_2, f), \quad \varepsilon = u, v, s \quad (15)$$

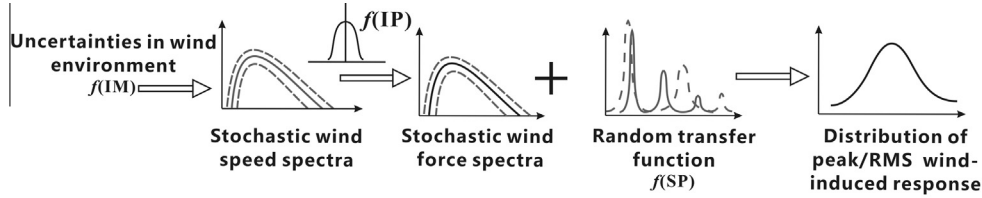
to those of the global floor forces as

$$S_{F_x, F_z}(f) = \int_0^H \int_0^H q_{z_1} q_{z_2} \lambda_{\alpha}^2 C_{\alpha}^2 \eta_{\alpha}^2 J_e(z_1) J_e(z_2) \varphi_{\alpha}(z_1) \varphi_{\alpha}(z_2) S_{ee}(z_1, z_2, f) dz_1 dz_2, \quad \alpha = x, y, \theta; \quad \varepsilon = u, v, s \quad (16)$$

Table 1

Characteristics of random variable (Intensity measure).

Variable	Distribution	Description	Mean value	COV	Unit
C_z	Lognormal	Exponential decay coefficient with respect to turbulence	8.0	0.1	–
$E(x)$	Normal	Estimated mean annual maximum wind speed	26.3	0.03	m/s
$S^2(x)$	Normal	Estimated variance of annual maximum wind speed	34.9	0.2	–
$E(x)$	Normal	Estimated mean daily maximum wind speed	11.6	0.02	m/s
$S^2(x)$	Normal	Estimated variance of daily maximum wind speed	18.7	0.07	–
–	Lognormal	Exposure constant	0.3	0.1	–
I_u	Lognormal	Turbulence intensity (along-wind)	$-0.055(z/H) + 0.1533$	0.17	%
I_v	Lognormal	Turbulence intensity (crosswind)	$0.88I_u(z)$	0.17	%

**Fig. 1.** Uncertainty propagation through the closed-form solution.

where q_{z1} and q_{z2} are the velocity pressures; $\lambda_x = \lambda_y = B$ and $\lambda_\theta = B^2$, in which $B = \sqrt{bd}$ is the building characteristic dimension (b and d are the planar dimensions of the building perpendicular and parallel to the along-wind direction, respectively); $c_{\alpha\epsilon}$ is the generalized aerodynamic force coefficient corresponding to excitation in the ϵ component and effect in the α direction, combined with the shape coefficient $\eta_{\alpha\epsilon}$; $J_\epsilon = \{2I_u, I_v, 1\}$, in which I_u and I_v are the along-wind and crosswind turbulence intensity, respectively. φ_α is the first order mode shape in the α direction. $c_{\alpha\epsilon}$ and $\eta_{\alpha\epsilon}$ are fitted in [50] based on wind tunnel tests and taken as the mean values of the random variables in Table 2. Together with the uncertainties involved in the intensity measure (IM) (with PDF $f(IM)$), these uncertainties of interaction phenomenon (IP) (with PDF $f(IP)$) propagate into the stochastic wind force spectra (Fig. 1). For each crisp realization, there are generally two types of approaches to evaluate the double integration in Eq. (16). One way is the conventional numerical integration. Another one is a recently proposed Monte Carlo based algorithm [8], which can accurately capture the stochastic excitation under relatively fine discretization degree (namely large samples). Considering we use a relatively coarse discretization, we compute the integration in the former way.

Afterwards, the response power spectrum and force spectrum are related by transfer function (Eq. (18)) as

$$S_{\alpha\alpha}(z, f) = \phi_\alpha^2(z) \frac{S_{F_\alpha F_\alpha}(f)}{m_\alpha^2} |H_\alpha(f)|^2, \quad \alpha = x, y, \theta \quad (17)$$

$$|H_\alpha(f)|^2 = \frac{1}{(2\pi f_\alpha)^4} \frac{1}{[1 - (f/f_\alpha)^2]^2 + [2(\zeta_\alpha + \zeta_{A\alpha})f/f_\alpha]^2}, \quad \alpha = x, y, \theta; \quad \zeta_{A\theta} = 0 \quad (18)$$

in which f_α , m_α^* , ζ_a and ζ_{Aa} are the fundamental frequency, modal mass, structural damping ratio and aerodynamic damping ratio in the α direction, respectively. Some of them are modeled as random variables in Tables 2 and 3. In this study, we use amplitude independent structural damping ratio for simplicity, while refined models are available in [55–57]. The aerodynamic damping ratio is fitted as the function of reduced mean wind speed U^* in [58] and its variability is induced by U^* . Aerodynamic damping ratio in torsional wind is ignored due to the absence of convincing model. The uncertainties in structural parameters (SPs) modeled by $f(SP)$ and aerodynamic damping ratio would result in random transfer function (Fig. 1).

Table 2

Characteristics of random variable (Interaction phenomenon).

Variable	Distribution	Description	Mean value		COV	Unit
			0 degree	90 degree		
C_{xu}	Lognormal	Aerodynamic force and shape coefficients	1.5	1.5	0.1	–
η_{xu}	Lognormal		1.08	0.88	0.1	–
C_{yv}	Lognormal		–1.13	–1.39	0.1	–
η_{yv}	Lognormal		0.84	0.94	0.1	–
$C_{\theta v}$	Lognormal		0.44	1.1	0.1	–
$\eta_{\theta v}$	Lognormal		0.75	1.58	0.1	–
C_{ys}	Lognormal		0.1	0.2	0.1	–
η_{ys}	Lognormal		1.31	1.31	0.1	–
$C_{\theta s}$	Lognormal		0.5	0.5	0.1	–
$\eta_{\theta s}$	Lognormal		1.5	0.67	0.1	–
ζ_{Ax}	–	Modal aerodynamic damping ratio	$0.000075U^{*2} - 0.00014U^* - 0.001$			%
ζ_{Ay}	–	Modal aerodynamic damping ratio	$\frac{0.0025(1 - (U^*/9.8)^2)(U^*/9.8) + 0.000125(U^*/9.8)^2}{(1 - (U^*/9.8)^2)^2 + 0.0291(U^*/9.8)^2}$			%
S_ϵ	Lognormal	Strouhal number	0.2		0.1	–

Table 3

Characteristics of random variable (structural parameter and resistance).

Variable	Distribution	Description	Mean value	COV	Unit
ξ_α	Lognormal	Modal damping ratio in α direction	1	0.15	%
f_α	Lognormal	Fundamental frequency in α direction	-	0.07	Hz
ρ_j	Truncated Normal	Joint action factor	0.85	0.1	-
R	Lognormal	Inter-story drift threshold (50 year)	1/400	0.1	-
R	Lognormal	Top deflection threshold (50 year)	1/500	0.1	-
R	Lognormal	Perception threshold (5 year)	$\exp(-3.65 - 0.41 \ln f_\alpha)$	0.1	m/s ²
R	Lognormal	Perception threshold (10 year)	0.25	0.1	m/s ²

- in the Mean value column implies that the fundamental frequencies are design dependent.

Then the peak value displacement and rotation responses are derived as

$$\sigma_\alpha(z) = \sqrt{\int_0^\infty S_{\alpha\alpha}(z, f) df}, \quad \alpha = x, y, \theta \quad (19)$$

Together with the mean component

$$\bar{u}(z) = q_z B C_{xu} \eta_{xu} \quad (20)$$

the along-wind most unfavorable peak displacement can be expressed as

$$u(z) = \bar{u}(z) + \sqrt{g_x^2 \sigma_x^2(z) + b^2 g_\theta^2 \sigma_\theta^2(z)/4} \quad (21)$$

The peak value factor is

$$g_\alpha = \sqrt{2 \ln f_\alpha T} + 0.5772 / \sqrt{2 \ln f_\alpha T}, \quad \alpha = x, y, \theta \quad (22)$$

in which f_α denotes the mean zero-crossing rate of the response process approximated by the fundamental frequency in the α direction, and T is the observation time duration. There is only fluctuating component in crosswind response as

$$v(z) = \sqrt{g_y^2 \sigma_y^2(z) + d^2 g_\theta^2 \sigma_\theta^2(z)/4} \quad (23)$$

In Eqs. (21) and (23), the RMS responses are further simplified into background and resonant component to reduce computational effort [13]. Though it is conventionally claimed that background component might be negligible for tall building design, the widely adjusted frequency ranges prevent us from doing this, hence the mean, background and resonant components are all taken into account in this study.

For the peak acceleration response, the partial dependence among the directional maximum acceleration components should be taken into account. Then the covariance between any two motion components in the sway and torsional directions cannot be omitted such that the peak resultant acceleration can be expressed as

$$a_{\text{peak}} = \sqrt{g_x^2 \sigma_x^2 + g_y^2 \sigma_y^2 + (b^2 + d^2) g_\theta^2 \sigma_\theta^2 / 4 + b r_{y\theta} g_y g_\theta \sigma_y \sigma_\theta - d r_{x\theta} g_x g_\theta \sigma_x \sigma_\theta} \quad (24)$$

in which $r_{y\theta}$ and $r_{x\theta}$ are the inter-component correlation coefficients. Since these coefficients are often unquantifiable, this expression is conventionally simplified as

$$a_{\text{peak}} = \sqrt{g_x^2 \sigma_x^2 + g_y^2 \sigma_y^2 + (b^2 + d^2) g_\theta^2 \sigma_\theta^2 / 4} \quad (25)$$

but it would result in too conservative and unrealistic results. Isyumov et al. [59] suggested a joint action factor ρ_j to represent the relationship between the maxima of independent response components as

$$a_{\text{peak}} = \rho_j \sqrt{g_x^2 \sigma_x^2 + g_y^2 \sigma_y^2 + (b^2 + d^2) g_\theta^2 \sigma_\theta^2 / 4} \quad (26)$$

In general the joint action factor ranges between 0.7 and 1 depending on the ratio of the smaller component to the larger component response, so the mean value 0.85 and truncated normal distribution are imposed on the random variable (Table 3).

Hereto, the displacement and RMS and peak acceleration responses would exhibit overall variability (Fig. 1) by considering the uncertainties involved in IM, IP and SP, which stem from basic variable and parametric uncertainty rather than a single wind process. When these uncertainties propagate through the closed-form solution to the performance measures, the resistances R in the serviceability limit states in the probabilistic constraints in (2)–(4) are also uncertain (Table 3). All the uncertainties propagate through the excitation–response–performance chain based on the joint PDF of the random variables in Tables 1–3 [19–23,60].

4. A many-objective formulation for PBWE of tall buildings

In the objective function (1) of RDO, there are twofolds of preferences in the total cost function, the first fold includes the loss coefficients and the second one includes the risk-taking preferences of the decision maker. All the design variable related items are aggregated into a single objective. Both initial cost (denoted as C_1) and the PFs include the design variable vector $\mathbf{x}$, hence RDO is inherently a multi-objective optimization problem (MOOP) using weighted-sum approach [61]. In the general formulation of constrained MOOP

$$\begin{aligned} &\text{Minimize : } \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_i(\mathbf{x}))^T \\ &\text{subject to : } g_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, k \end{aligned} \quad (27)$$

there is no universal optimum for all the objectives. Rather, there exist a set of solutions and each one of them is appealing in some aspects, so the common optimality condition used in mono-objective optimization is replaced by Pareto optimality. Given m objectives and any two design vector $\mathbf{x}_1$ and $\mathbf{x}_2$ satisfying

$$\begin{aligned} &\forall i \in (1, 2, \dots, m), [f_i(\mathbf{x}_1) \leq f_i(\mathbf{x}_2)], \\ &\exists j \in (1, 2, \dots, m), [f_j(\mathbf{x}_1) < f_j(\mathbf{x}_2)] \end{aligned} \quad (28)$$

$\mathbf{x}_1$ dominates $\mathbf{x}_2$. If and only if there is no design vector $\mathbf{x}$ belonging to $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) \in \mathbf{R}$ such that $\mathbf{x}$ dominates $\mathbf{x}_1$, $\mathbf{x}_1$ is a non-dominated solution in $\mathbf{R}$. All the solutions like $\mathbf{x}_1$ constitute the Pareto optimal set. The mapping of Pareto optimal set in the objective space is termed Pareto front.

However, the weighted-sum formulation can only handle rigorously convex Pareto front, and fails in dealing with non-convex Pareto front [61]. RDO might be such a complex problem and the assumption of a convex Pareto front is far from rigor. The existences of non-convex Pareto front for multi-objective optimization involving numerical characteristics of random variables have been reported in [62,63]. On the other side, uncertainties in loss coefficient possess an indispensable portion [64], and this is the very case for the loss incurred by NSE damage and habitability violation. If C_{1j} , C_{2k} and C_l are assigned with subjectively selected values,

excessive uncertainties would be introduced but never quantified. More importantly, the number of failure modes in practical RDO problems would be relatively large. If there are n failure modes, their dominance relationships with the C_1 are established in the $n + 1$ dimensional objective space, in which the total errors of loss estimation accumulate in the form of exponentially increasing. This observation, which has not been explicitly addressed in existing RDO literature, implies that the adverse effects of model uncertainty on the additive model are much more significant than they appear to be, and subsequently, pose great challenge on RDO methodologies, hence new RDO formulation is required.

In light of the available amount of information on NSE loss, RDO can be formulated as multi-objective RBDO (MORBDO) [65] in two different cases. If the damage loss of NSEs can be listed in detail, the performance losses are aggregated into an expected loss item. The objective vector consists of initial cost, expected loss of NSEs and PFs of habitability performance. C_T is transformed into an objective vector.

$$\mathbf{f} = \left(C_1, \sum_{j=1}^{N+1} Pf_{Disj} C_{1j}, Pf_{Peak}, Pf_{RMS} \right) \quad (29)$$

in which Pf_{Peak} and Pf_{RMS} respectively denote the atop peak and the most unfavorable RMS acceleration response and $N + 1$ is the aforementioned number of independent failure modes associated with NSEs. In the other case, loss estimation for NSEs is not available. Then the minimization of NSE loss is further decomposed into the minimization of all the corresponding PFs, and the objective vector consists of initial cost, PFs of drift and habitability performances. The dimensionality of objective space increases to $N + 4$ as

$$\mathbf{f} = (C_1, Pf_{Dis1}, Pf_{Dis2}, \dots, Pf_{Dis(N+1)}, Pf_{Peak}, Pf_{RMS}) \quad (30)$$

When the dimensionality of the objective space is larger than three, the MOOP is termed many-objective optimization problem [66,67]. Though there are respectively 4 and $N + 4$ objectives in (29) and (30), the corresponding Pareto fronts are unnecessary to be with the same dimensionality. Based on the concept of collinearity [63], we know two objectives f_1 and f_2 are locally collinear at point $\mathbf{x}$ if $\nabla f_1(\mathbf{x}) = \beta \nabla f_2(\mathbf{x})$ conditioned under $\beta > 0$. Otherwise, they are locally conflicting. While these two objectives are globally conflicting in the feasible design space, if the two minimization optimization problems with mono-objective f_1 and f_2 have different global optimal solutions. In light of this, any two objectives f_1 and f_2 , which are locally collinear or highly nearly collinear at the proximity of the optimal Pareto front, can be regarded as non-conflicting hence one of them is redundant. Assuming the Pareto optimal set consists of k solution points, any one of f_1 and f_2 is considered redundant to construct their conflicting relationships with any other objectives, when f_1 and f_2 are locally collinear at all the k solution points. In this way, the true dominance structure of all the $N + 4$ objectives has a high inclination to live in a much lower dimensional objective space at the global optimum locations.

Considering the optimal Pareto front might never be obtained in practice, we seek to eliminate the redundant objectives at the nearly optimal positions of the objectives. Two subsequently imposed questions are: how to detect these redundant objectives and how to hit the nearly optimum solutions of a specific problem. Though Tchebycheff approach [68] aided with smoothing function [69] and boundary intersection approach [68] can effectively deal with non-convex Pareto front, gradient-based methods would be computationally burdensome and low efficient on high-dimensional objective space when we solve the MOOP without preference. That means we have to sequentially solve a set of mono-objective problems with high-dimensional weight vectors. Without a doubt, a considerable portion of the mono-objective problems have identical solution owing to collinearity of objec-

tives. Multi-objective evolutionary algorithm (MOEA), however, can efficiently approximate the Pareto front within a single run [61], and exhibits extra competence on the studied problem with discrete variables and nonlinear probabilistic constraints. Once the approximate Pareto front is derived, the original multi-objective optimization and decision making is simplified as a multi-attribute decision making [70].

5. Solving RDO using Micro-MOPSO-PCA

We propose a high efficient Micro-MOPSO-PCA algorithm to meet three requirements of MORBDO on MOEA. First, it is based on up-to-date metaheuristics to enhance efficiency; second, the population size is very small hence only a small number of function evaluations are required to approximate the Pareto front satisfyingly; Last but not least, it is capable of handling many-objective problems. There are several available Micro-MOEAs, such as Micro-MOGA [71,72] and Micro-MOPSO [73]. Among existing multi-objective metaheuristics, MOPSO algorithms exhibit higher convergence rate thus a significant saving in computational time can be expected. Because the initially proposed Micro-MOPSO is an unconstrained version and penalty-based constraint handling techniques are problem dependent [74], we equip it with a simple but efficient constraint handling technique [75].

The standard PSO algorithm is initialized with a set of particles and each one of them is iteratively moved across the search space and is attracted to the position with the best fitness historically achieved by the particle itself and by the best among the neighbors [76]. It requires no gradient information and uses only primitive mathematical operators and two equations as

$$\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + \mathbf{v}_i(t+1) \quad (31)$$

$$\mathbf{v}_i(t+1) = w\mathbf{v}_i(t) + \mathbf{r}_1 c_1 (\mathbf{x}_L - \mathbf{x}_i(t)) + \mathbf{r}_2 c_2 (\mathbf{x}_G - \mathbf{x}_i(t)) \quad (32)$$

where $\mathbf{v}_i(t)$ and $\mathbf{x}_i(t)$ denote the velocity vector and the position vector of the i th particle at iteration t , respectively. $\mathbf{x}_L$ is the best solution found by the i th particle itself at the current iteration, and $\mathbf{x}_G$ is the global best solution found among the entire swarm up to the current iteration. The acceleration coefficients c_1 and c_2 represent “trust” settings which indicate the degree of confidence in the best solution found by the individual particle and by the whole swarm, respectively. $\mathbf{r}_1$ and $\mathbf{r}_2$ are two vectors containing random numbers uniformly distributed in the interval [0,1] to encourage the exploration. When any particle violates the upper or lower boundary constraints, a recombination operator proposed in [77] is applied to guarantee that its value is always updated in the feasible design space.

This standard PSO strategy must be modified to solve MOOPs. As illustrated in Section 3, the solution set of a problem with multiple objectives consists of a set of solutions rather than a single solution. Then there are three main goals to achieve: maximizing the number of solutions in the Pareto front; minimizing the distance of the approximate Pareto front with respect to the global Pareto front and maximizing the spread of solutions [78]. Though the failure probability objectives in the MORBDO problem are well bounded within [0, 1], on-line normalization [79] is still mandatory to prevent solution clustering owing to the scaling of the failure probability associated with different failure mode. Besides, we handle discrete variables in the way proposed in [80].

In Section 3, the original mono-objective problem is transformed into a many-objective version under two cases, in which 4 and $N + 4$ objectives are involved, respectively. Since the number of N is usually large, the both cases, especially the latter, are very challenging to solve because many-objective problems would pose three additional challenges [67] to MOEAs. First, the number of

solutions required to approximate a high-dimensional Pareto front dramatically increases; Second, many objectives would make each solution appealing in at least one aspect and the performance of MOEA would be greatly impaired based on conventional dominance structure; Last, the high-dimensional Pareto front is inconvenient for visualization hence decision-making. MOPSO can efficiently deal with two or three objectives, but its performance on high-dimensional objective space has not been widely tested. Existing approaches for many-objective evolutionary algorithms can be categorized into: dimensionality reduction, preference incorporation, selection pressure enhancement and different fitness evaluation schemes [81–85]. Among them, dimensionality reduction method is particularly suitable for solving RDO problem owing to the possibility of performing objective screening [84,85].

Regarding each objective as a random variable, we have each candidate solution as an objective realization. PCA is employed to seek for a less redundant subset of this set of random variables by retaining as much variation of the data set as possible. The efficacy of PCA combined with other MOEA has been tested in [85]. We apply the kernel-learning based nonlinear dimensionality reduction and integrate it with Micro-MOPSO. Details of the objective screening strategy are described in [85]. Once the non-redundant objectives are identified, the order-reduced correlation matrix is used to further reduce the dimensionality. As validated in [85,86], ϵ -dominance relationship can handle many-objective problem based on conventional dominance relationship, we apply ϵ -dominance on the auxiliary and final archive of the Micro-MOPSO algorithm. So the number of final solutions is directly related with the number of grids and its impact on the efficacy of PCA is examined in the next section.

The overall three-loop procedure of serviceability PBWE of tall buildings using many-objective RBDO is established and shown in Fig. 2. In the inner loop, we use uncertainty propagation techniques to derive the PFs corresponding to each candidate solution. In the mid loop, the Micro-MOPSO-PCA algorithm seeks for an approximate Pareto front conditioned under a fixed number of objectives. While in the outer loop, it seeks for a possibly reduced number of objectives. The optimization run is carried out iteratively until no dimensionality reduction is available. Control parameters of the Micro-MOPSO algorithm are listed in Table 4.

6. Illustrative example

To validate the proposed RDO methodology, used in this study is a 45-story tall steel frame subject to 3D wind loads with two incident direction cases. Its geometric shape is identical to the CAARC benchmark [87]. Its plan view and elevation view are depicted in Fig. 3, where C1 to C6 respectively represent the grouped column lines while all the beam members at the same floor are grouped together with an identical cross sectional size. A series of K-bracings are implemented to the central bays and two sets of belt trusses are established at the two refuge-and-mechanical floors. All the member sizes are selected in accordance with AISC code [88], and the unit cost of steel material $C_U = 3700$ USD/t. A preliminary strength check is performed and the corresponding lower bounds of the element sizes and fundamental frequencies are listed in Table 5. Rigid beam-to-column connections are assumed while the belt trusses and K-bracings are simply connected.

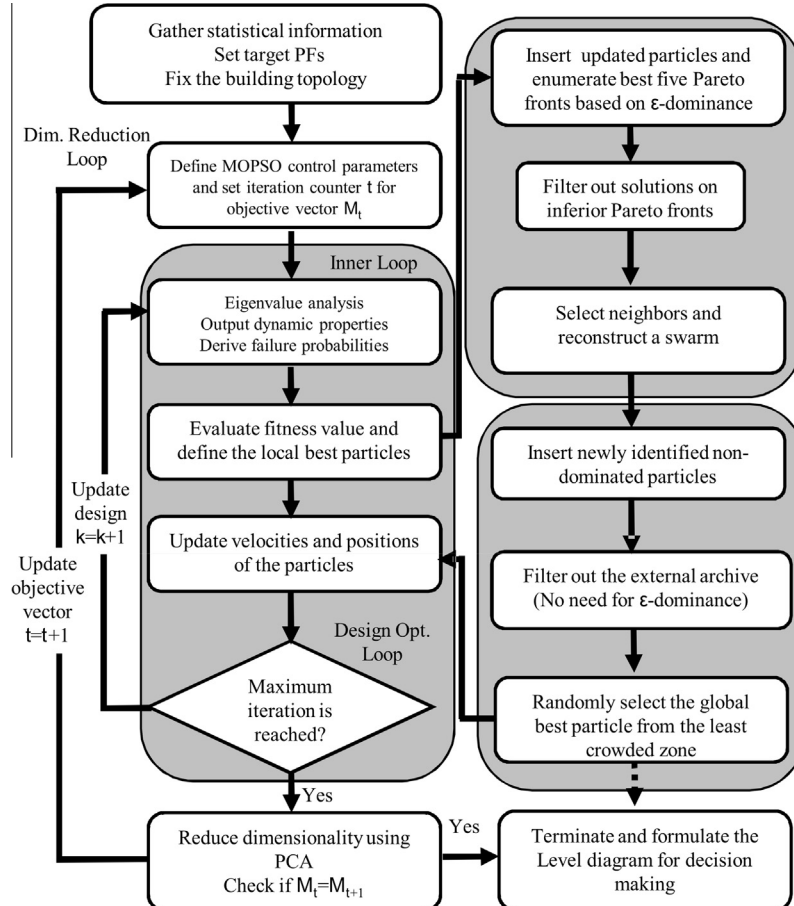


Fig. 2. Flowchart of the Micro-MOPSO-PCA algorithm.

Table 4
Micro-MOPSO control parameter settings.

Parameter	Value
Population size	5
Maximum generation	1000
Inertia weight w	Uniformly distributed in [0,1]
Cognitive parameter c_1	1.8
Social parameter c_2	1.8
Auxiliary archive size	100
Final archive size	100 (no need for ε dominance)
Reinitializing generation	100
Mutation mechanism/probability	Polynomial/0.1

All the joints on the same floor level move together in a plane based on the rigid diaphragm assumption.

6.1. Deriving PFs using Monte Carlo simulation (MCS) and Latin hypercube sampling (LHS)

We all know that the accuracy of probabilistic analysis in conventional double-loop RBDO formulation is crucial to the overall convergence and this importance is further elucidated in [89]. Some classical uncertainty propagation techniques, such as first order reliability method and second order reliability method, might be unreliable when encountering problems with more than 20

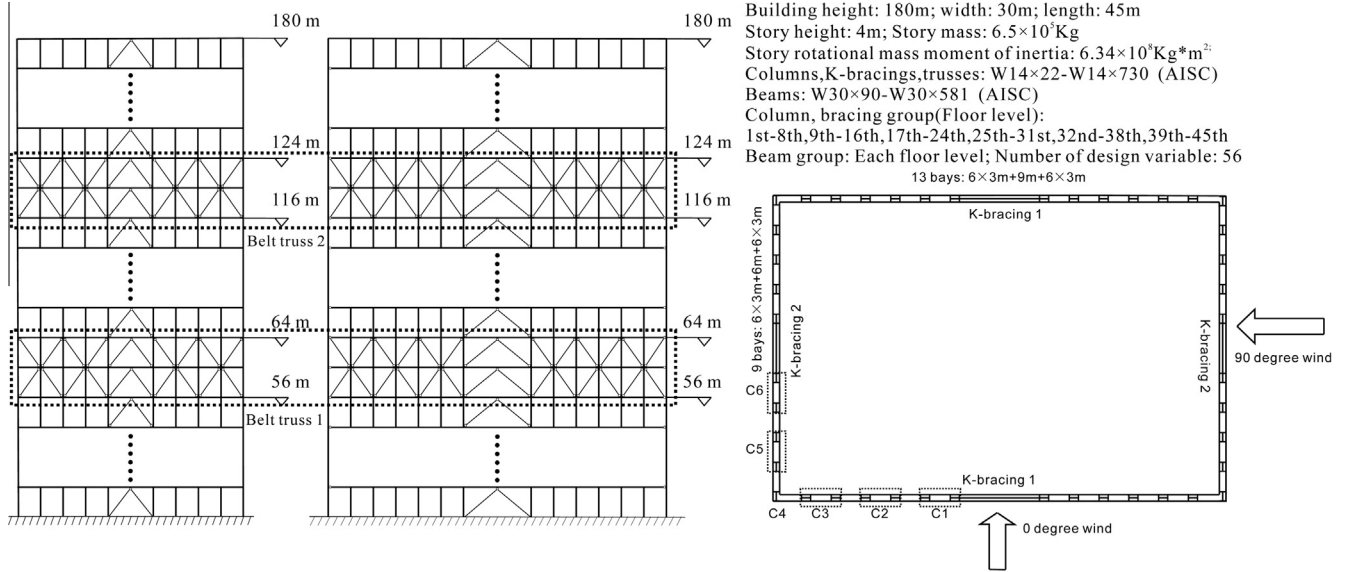


Fig. 3. 45-story wind-excited benchmark tall steel frame.

Table 5
Initial design for verification.

Story No.	1–8	9–16	17–24	25–31	32–38	39–45
Column	W14 × 500	W14 × 455	W14 × 342	W14 × 257	W14 × 193	W14 × 145
Beam	W30 × 391	W30 × 357	W30 × 326	W30 × 261	W30 × 211	W30 × 191
K-bracings	W14 × 211		Belt-truss 1	W14 × 211	Belt-truss 2	W14 × 211
Frequency	Mode 1 (Sway) 0.195 Hz		Mode 2 (Sway) 0.229 Hz		Mode 3 (Torsional) 0.475 Hz	

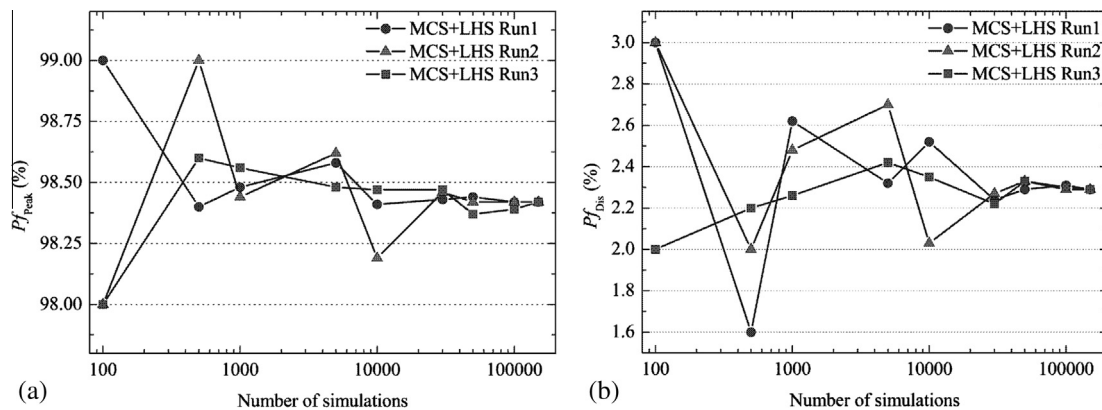


Fig. 4. Required number of Latin hypercube samples. (a) PF of resultant atop acceleration; (b) PF of atop deflection.

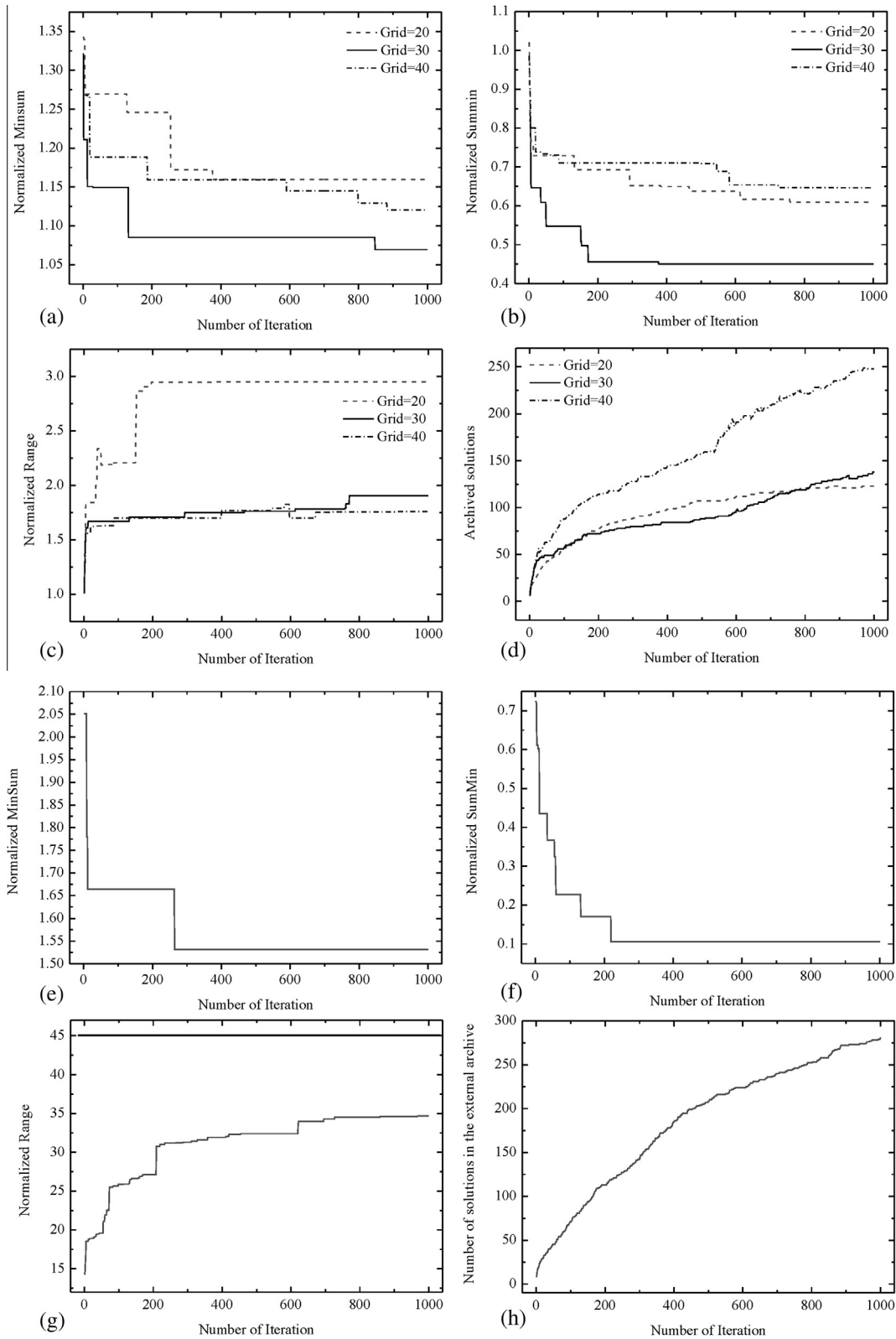


Fig. 5. Convergence history. (a–d) 4-objective problem; (e–h) 49-objective problem.

random variables [90], and this is the case for the studied RDO problem. So we use MCS in conjunction with LHS [91] in this work. In order to examine the performance of MCS + LHS on computing the PFs, carried out first is a parametric study that monitors the influence of the number of simulation runs on the accuracy of performance indicator (resultant atop acceleration and atop along-

wind displacement) using the element size combination in Table 5 without vibration control. Three sets of 150,000 samples are initiated and different resampled realizations are drawn from each set with gradually increasing sample size. In Fig. 4, it is evident that though LHS is implemented, 100,000 samples are required to derive convincing PFs for both displacement and acceleration

response in all three runs, and this number can be deemed as a minimum resolution to differentiate the objectives. For high-dimensional stochastic problem, the sample size required to compare the probabilistic performance indicators dramatically increases. Closed-form solution can release the overall computational burden, but it is not trivial owing to the numerical integration for computing the background component of the response.

6.2. Objective dimensionality reduction

Considering the monotonous increasing of STMD cost, we use $\gamma = 0.5$ as a typical value to perform objective dimensionality reduction and the maximum generation is predefined as 1000 for

Table 6
Principle components for 4-objective problem.

	PC ₁	PC ₂	PC ₃
<i>(a) Grid = 20</i>			
C_1	−0.8637	−0.0164	0.0611
Pf_{RMS}	0.3470	−0.0980	0.7874
Pf_{Peak}	0.2652	−0.6441	−0.5146
NSE loss	0.2514	0.7584	−0.3340
Cum. Contr.	77.44%	95.52%	100.00%
<i>(b) Grid = 30</i>			
C_1	0.7810	−0.3331	0.1704
Pf_{RMS}	0.0823	0.8363	−0.2092
Pf_{Peak}	−0.3588	−0.4290	−0.6612
NSE loss	−0.5044	−0.0742	0.7000
Cum. Contr.	59.12%	90.58%	100.00%
<i>(c) Grid = 40</i>			
C_1	−0.8358	−0.0018	0.2267
Pf_{RMS}	0.1995	0.7615	−0.3611
Pf_{Peak}	0.1461	−0.6364	−0.5689
NSE loss	0.4902	−0.1233	0.7032
Cum. Contr.	59.02%	92.28%	100.00%

Table 7
Correlation matrix for dimensionality reduction.

	C_1	Pf_{RMS}	Pf_{Peak}	NSE loss
<i>(a) Grid = 20</i>				
C_1	1.0000	−0.8931	−0.7512	−0.7602
Pf_{RMS}	−0.8931	1.0000	0.7796	0.6801
Pf_{Peak}	−0.7512	0.7796	1.0000	0.4168
NSE loss	−0.7602	0.6801	0.4168	1.0000
Overall Contr.	60.22%	12.28%	14.13%	13.37%
<i>(b) Grid = 30</i>				
C_1	1.0000	−0.8080	−0.4758	−0.2927
Pf_{RMS}	−0.8080	1.0000	0.5998	0.0430
Pf_{Peak}	−0.4758	0.5998	1.0000	0.1259
NSE loss	−0.2927	0.0430	0.1259	1.0000
Overall Contr.	39.83%	22.82%	17.52%	19.83%
<i>(c) Grid = 40</i>				
C_1	1.0000	−0.8784	−0.3323	−0.5116
Pf_{RMS}	−0.8784	1.0000	0.4996	0.2953
Pf_{Peak}	−0.3323	0.4996	1.0000	0.1605
NSE loss	−0.5116	0.2953	0.1605	1.0000
Overall Contr.	41.62%	22.64%	17.23%	18.51%

Table 8
Loss functions for non-structural elements.

Function	Loss (USD)
Uniform	1,000,000
Linear	41,000 * N
Quadratic	−3874 * N ² + 182,000 * N − 338,700

N = Story number.

the Micro-MOPSO algorithm. In each run, 5000 candidate solutions are evaluated and identical times of eigenvalue analyses are conducted during the whole optimization process for a fixed objective vector. Based on the design variable set, the design space is comprised of 3.65×10^{16} combinations while the stochastic algorithm only explores a very small portion (1.37×10^{-13}) of it to locate a high quality approximation of the global Pareto front.

Since the number of floor levels is 45, the objective numbers for the studied RDO problem are 4 and 49 in the two cases, respectively. For the 4-objective problem, we first examine the influence of the number of archived solutions on dimensionality reduction. Based on ε -dominance, each objective is partitioned into 20, 30 and 40 parts in the normalized objective space. Each case is

Table 9
Collinearity between NSE loss and Max(Pf_{Dis}).

<i>(a) PCA (Uniform)</i>					
	PC ₁	PC ₂	PC ₃	PC ₄	
C_1	−0.8916	−0.0468	−0.0480	0.0244	
Pf_{RMS}	0.2848	−0.5680	−0.6212	−0.1020	
Pf_{Peak}	0.2029	−0.3940	0.7769	−0.0095	
NSE loss	0.2270	0.4299	−0.0896	0.7454	
Max(Pf_{Dis})	0.1769	0.5788	−0.0181	−0.6583	
Cum. Contr.	69.63%	90.98%	98.74%	100%	
<i>Correlation matrix (Uniform)</i>					
	C_1	Pf_{RMS}	Pf_{Peak}	NSE loss	Max(Pf_{Dis})
C_1	1.0000	−0.8280	−0.5930	−0.6172	−0.5438
Pf_{RMS}	−0.8280	1.0000	0.7156	0.5708	0.4398
Pf_{Peak}	−0.5930	0.7156	1.0000	0.6321	0.5517
NSE loss	−0.6172	0.5708	0.6321	1.0000	0.9446
Max(Pf_{Dis})	−0.5438	0.4398	0.5517	0.9446	1.0000
Overall Contr.	55.42%	15.54%	10.86%	8.3%	9.88%
<i>(b) PCA (Linear)</i>					
	PC ₁	PC ₂	PC ₃	PC ₄	
C_1	0.7090	−0.5429	−0.0305	0.0410	
Pf_{RMS}	−0.4996	−0.2862	−0.5351	−0.4268	
Pf_{Peak}	−0.4606	−0.2362	0.6042	0.4086	
NSE loss	0.0807	0.5404	−0.4358	0.5582	
Max(Pf_{Dis})	0.1706	0.5249	0.3971	−0.5811	
Cum. Contr.	56.15%	91.00%	97.80%	100.00%	
<i>Correlation matrix (Linear)</i>					
	C_1	Pf_{RMS}	Pf_{Peak}	NSE loss	Max(Pf_{Dis})
C_1	1.0000	−0.8204	−0.7490	−0.3626	−0.2226
Pf_{RMS}	−0.8204	1.0000	0.7783	0.0550	0.1385
Pf_{Peak}	−0.7490	0.7783	1.0000	0.0559	0.0531
NSE loss	−0.3626	0.0550	0.0559	1.0000	0.8309
Max(Pf_{Dis})	−0.2226	0.1385	0.0531	0.8309	1.0000
Overall Contr.	38.51%	19.22%	16.71%	12.52%	13.05%
<i>(c) PCA (Quadratic)</i>					
	PC ₁	PC ₂	PC ₃	PC ₄	
C_1	−0.8925	−0.0028	−0.0558	−0.0188	
Pf_{RMS}	0.1931	−0.7055	0.5117	0.0557	
Pf_{Peak}	0.2655	−0.0005	−0.4101	−0.7492	
NSE loss	0.1889	0.7087	0.5090	0.0548	
Max(Pf_{Dis})	0.2450	0.0000	−0.5548	0.6574	
Cum. Contr.	81.78%	92.80%	99.21%	100.00%	
<i>Correlation matrix (Quadratic)</i>					
	C_1	Pf_{RMS}	Pf_{Peak}	NSE loss	Max(Pf_{Dis})
C_1	1.0000	−0.8730	−0.7548	−0.7443	−0.8204
Pf_{RMS}	−0.8730	1.0000	0.7996	0.7982	0.9708
Pf_{Peak}	−0.7548	0.7996	1.0000	0.6278	0.7756
NSE loss	−0.7443	0.7982	0.6278	1.0000	0.7748
Max(Pf_{Dis})	−0.8204	0.9708	0.7756	0.7748	1.0000
Overall Contr.	65.16%	10.22%	7.29%	10.12%	7.22%

performed 5 times and the ones with the best convergence are illustrated. In the three cases, 123, 138 and 249 solutions are archived respectively. In contrast to the mono-objective optimization, the continuing decreasing of a particular objective (for a minimization problem) is no longer a competent convergence measure in multi-objective cases, because the convergence is measured towards the Pareto front rather than a single solution point. Assuming ψ is the current approximation set of the Pareto front, we have all the K objective values of each solution point in this set. Then the convergence is measured using four metrics [84], i.e., distance of the approximation set with respect to the central region of global Pareto front

$$\text{Minsum}(\Psi) = \min_{x \in \Psi} \sum_{i=1}^K f_i \quad (33)$$

that with respect to the edge of global Pareto front

$$\text{Summin}(\Psi) = \sum_{i=1}^K \min_{x \in \Psi} f_i \quad (34)$$

spread of solutions in the objective space

$$\text{Range}(\Psi) = \sum_{i=1}^K (\max_{x \in \Psi} f_i - \min_{x \in \Psi} f_i) \quad (35)$$

and the number of solutions in the approximation set. All the results are obtained in a normalized objective space, so the theoretical upper bounds of the first three metrics are exactly the number of objectives. The convergence of the 4-objective problem measured using these four metrics are shown in Fig. 5(a)–(d). In all the three cases, gradual convergence is observed. A grid number of 30 may be more appropriate as it exhibits better overall convergence towards

either the central region or all the edges of the Pareto front, though the range and the number of solutions it archives is less attractive. We also see that 1000 is suitable as a maximum generation to detect the generation after which the approximate degree no longer improves significantly.

As shown in Table 6, three principle components are identified via eigenvalue decomposition of the learned kernel matrix for all the three cases, in which different numbers of grids are used for ε -dominance. Except the second case, PCA alone cannot reduce the dimensionality of the objective vector. With a deep look into the correlation matrices (Table 7), we find Pf_{RMS} , Pf_{Peak} and NSE loss can be regarded as globally collinear, at least in the proximity of the approximate Pareto front. Hence only one of them is retained. Because Pf_{RMS} carries more overall variability than the other two objectives, the global conflicting relationship is established between C_1 and Pf_{RMS} . The number of solutions archived has trivial impact on dimensionality reduction, hence the rationality of assigning grid number as 30 is reconfirmed.

For the 49-objective problem, the number of objectives is first reduced from 49 to 13 based on PCA. Then we construct the reduced correlation matrix and the final non-redundant objectives are C_1 , Pf_{RMS} , $Pf_{\text{Dis}7}$, $Pf_{\text{Dis}12}$, $Pf_{\text{Dis}18}$, $Pf_{\text{Dis}26}$ and $Pf_{\text{Dis}45}$ (The numbers correspond to the floor level number). From results of repeated runs, we find that though C_1 and Pf_{RMS} are always non-redundant, the non-redundant objectives associated with displacement response are not fixed in different runs. In this sense, though the convergence histories plotted in Fig. 5(e)–(h) are satisfying, the uncertainties induced by the stochastic search process of the algorithm prevent us from obtaining the true dominance structure and decision making on such an uncertain high-dimensional objective space is prohibitive.

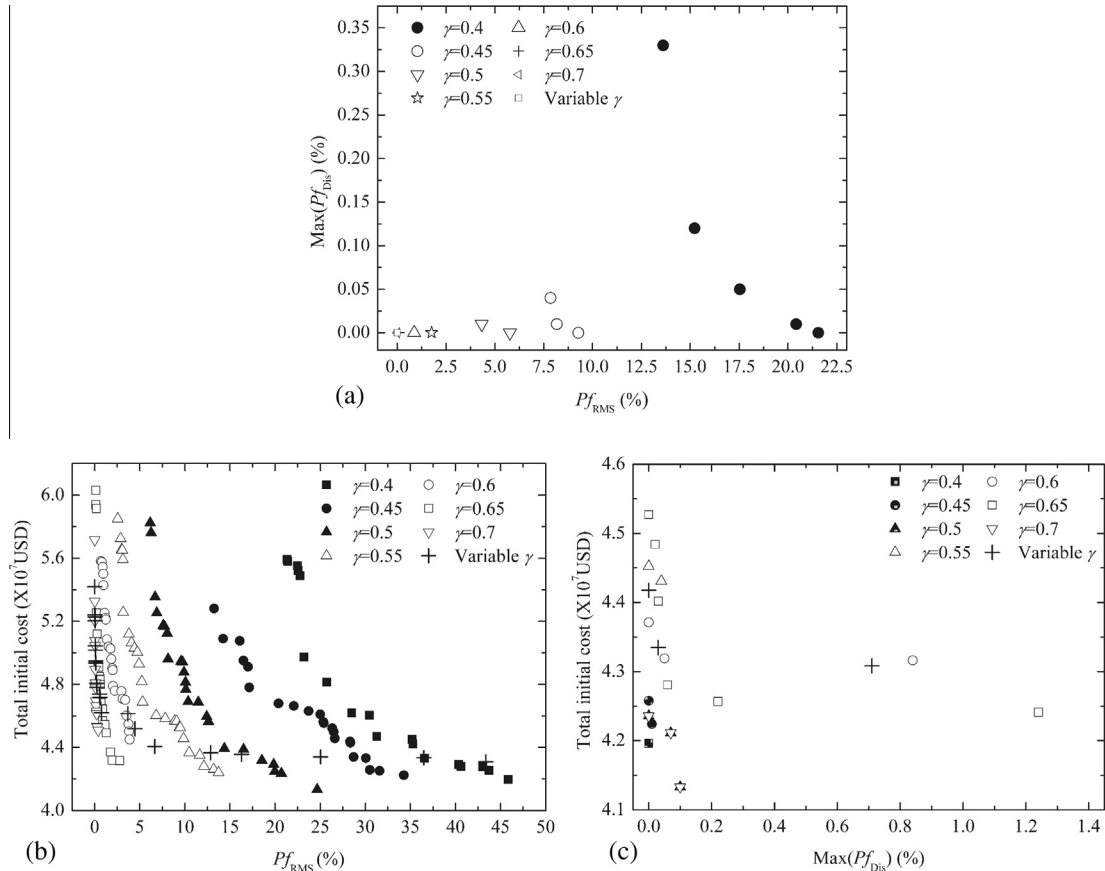


Fig. 6. Approximate sets under different ARLs. (a) Collinearity between the $\text{Max}(Pf_{\text{Dis}})$ and Pf_{RMS} ; (b) Total initial cost and Pf_{RMS} ; (c) Total initial cost and $\text{Max}(Pf_{\text{Dis}})$.

To overcome this difficulty, we use the most unfavorable displacement $\text{Max}(P_{f_{\text{Dis}}})$ as a substitution for NSE loss. If we could obtain the same dominance relationship among the objectives by using either estimated NSE loss or $\text{Max}(P_{f_{\text{Dis}}})$, the conflicting relationships among all the other objectives can be readily revealed using additional stochastic analyses after the optimization process. To approve this approach, we employ three different types of loss functions (Table 8) for inter-story drift ratio, and the loss coefficient associated with the atop deflection response is assigned as 5×10^5 USD. In Table 9, both PCA and correlation matrix analyses reveal that $\text{Max}(P_{f_{\text{Dis}}})$ is an appropriate substitution of NSE loss. Once again, $P_{f_{\text{RMS}}}$ contributes most to the overall variability conditioned under all the NSE loss functions, so the final approximate Pareto front can be derived by solving the bi-objective problem including C_1 and $P_{f_{\text{RMS}}}$.

6.3. Bi-objective stiffness and vibration control optimization

Once the non-redundant objectives are identified, the final solutions are obtained by performing a bi-objective (C_1 and $P_{f_{\text{RMS}}}$) optimization. For the dynamic system, adjustments in frequencies and mode shapes are expected to reduce translational and torsional response. We calibrate the collinearity between the optimal acceleration and displacement response derived via the aforementioned dimensionality reduction algorithm. As an alternative of the impossible analytical method, we perform the bi-objective optimization under different ARLs. Fig. 6(a) shows the collinearity between the most unfavorable RMS acceleration $P_{f_{\text{RMS}}}$ and maximum displacement response $\text{Max}(P_{f_{\text{Dis}}})$. $\gamma = 0.5$ can be considered as a transition point. When γ is larger than 0.5, the bi-objective optimization converges to a single point and the collinearity between them is confirmed. While γ ranges from 0.4 to 0.5, though only involving several solutions, an approximation set is retainable, then we can conclude that $P_{f_{\text{RMS}}}$ and $\text{Max}(P_{f_{\text{Dis}}})$ are not collinear under relatively low ARLs. This might be due to the more contribution of sizing design variables on suppressing the response. Under these circumstances, the original problem cannot be solved in a bi-objective strategy but should be solved by optimizing all the three objectives (i.e., C_1 , $P_{f_{\text{RMS}}}$ and $\text{Max}(P_{f_{\text{Dis}}})$) simultaneously to represent the dimensionality of the true Pareto front.

When bi-objective optimization is performed, the conflicting relationships between total initial cost and $P_{f_{\text{RMS}}}$ under different ARL are shown in Fig. 6(b). Moderate ARL provides the widest spectra radius of the approximate Pareto front. Once ARL exceeds 0.6, RMS acceleration response is restricted to a narrow band. Solutions on the approximate Pareto front are distributed more uniformly within the ARL range of 0.45 to 0.65 and more solutions are obtained. In Fig. 6(c), $\text{Max}(P_{f_{\text{Dis}}})$ almost all approach zero and NSE loss is absent under low ARLs. While under high ARLs, stiffness requirement to diminish acceleration response is relaxed, so more trade-off pairs are found. With a variable γ , concurrent stiffness and vibration control optimization is performed as well. Rather than providing a usable approximate set, it offers the designer a bird-eye view on the available solutions high efficiently. Though a tri-objective optimization might be necessary under low ARLs, it is not considered in this study owing to its less attraction in minimizing the total initial cost.

7. Concluding remarks

In this work, an automated PBWE framework has been proposed for serviceability design of tall buildings under 3D wind loads, in which the elements of PBWE, including aerodynamic analysis, structural analysis and risk analysis, are incorporated in a hybrid stiffness and vibration control RDO formulation. The 3D

closed-form solution in frequency domain is used for uncertainty quantification and propagation. Particularly, we point out the multi-objective optimization nature of RDO, the irrationality of its mono-objective version for PBWE of tall buildings and propose a novel many-objective formulation for it. To high efficiently solve the many-objective RBDO problem, ϵ -dominance relationship and kernel-learning based PCA are employed to assist a constrained Micro-MOPSO with a very small population size. Thanks to the MOR-BDO formulation and Micro-MOPSO-PCA algorithm, the difficulties stemming from the absence of business loss model are overcome by identifying the lower dimensional trade-off pairs contributing most to the overall variability. The bi-dimensional Pareto front well approximates the dominance structure corresponding to the original high-dimensional objective vector. Once the approximate Pareto front is obtained, the original complex multi-objective decision making is simplified as the multi-attribute decision making, which can be performed readily and flexibly.

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